

Rats and their Control

The history of rats is as old as human-beings. In 'Bhagwat Puran', an old literature of 10th century A.D., people had been warned to desert their dwellings as soon as dead rats were seen. The existence of crop-pests like birds, rats, monkeys, etc. have been mentioned since 'Pre-Vadic' period and according to 'Kamandak' the heavy rain-fall, heavy drought, rats, locusts and parrots are serious problem for the crops and thereby economy of nation. Lord Ganesh is always conciliated before any worship in Hindu religion for he has subjugated the evil, i.e., a rat which is his convergence. Near Bikaner, in a temple at 'Deshnokh' 80 kg grain is given daily for the feeding of rats.

Rats have been the cause of severe damage to crops, fruits, trees, vegetables in the field and almost every food grain in houses and godowns. Srivastava (1968) has reported 110 species of rats, from India, damaging the crops. They are very active, cunning and prolific breeders so their population is increasing day-by-day. From a data of FAO and UNO it was calculated that world population of rats is more than 42,500 crores. The rat population in India is reported to be more than 10,000 millions. It is our surmise that if rats are allowed to damage freely it would be difficult for human-beings to live in their dwellings and nothing would be left for feeding of human-beings.

For the control of rats, it is essential to have a clearcut idea about their species, habit and habitat, mode of burrowing, their movements, preference of food and a detailed life history of different kinds of rats.



Fig. 1. A common rat, *Rattus rattus*.

Species of Rats

According to Srivastava (1968) 110 species are found in India of which 70 species belong to the family Muridae and order-Rodentia. The important ones belong to 2 sub-families viz., Murinae and Gerbilinae. Three types of rats are reported i.e., field rats, house rats and rats damaging in houses and field both. Only few important species of rats are given below —

1. *Rattus rattus* Linn. (House rat). It is found dwelling in houses, warehouses, packing houses, seed stores, ships and sea ports. This species is arboreal or climbing in nature and rarely form their burrows for living. Sixteen species are found causing enormous damage to coconuts. They breed throughout the year and the female is provided with 10 to 12 mammae for feeding of young ones. The tail of this rat is longer than the rest of the body. Two races of this rat are observed in India, one with reddish brown dorsal surface and other with black reddish dorsal surface of the body (Fig. 1).

2. *Tetra indica hardwicke* Gray (Indian gerbil). This is found in uncultivated field and is an important carrier of plague. The body colour is light brownish dorsally and darker with pale hairs on ventral side. The hind feet are much longer and well adapted for jumping. It damages grains, succulent stem, leaves, roots, fruits, particularly sorghum and millet crops.

3. *Rattus norvegicus* Berkenhout (Norway rat). This is mainly found in temperate regions and inhabits sea-shore cities. Actually, this is house rat and can live in the field, feeding on crops, also. Fur is softer than that of black rat and the grooved bristles are more slender and smaller in number. The body colour is dark-brown on the dorsal side while white coloured on the ventral side. This rat can also breed throughout the year and the female has 12 mammae.

This rat is larger and more robust than black rat. Tail is slightly shorter than the body length.

4. *Mus musculus* Linn (Albino rat). It inhabits the houses but sometimes goes to the gardens and crop fields near villages or towns. The body colour is dark to light brown dorsally and pale brown ventrally. This rat is very much active and can feed upon anything which comes in its way. This can breed throughout the year and the female has 12 mammae. Tail is shorter than the body. The most important thing for this rat is that it is used for scientific studies in laboratories for experimental purposes.

5. *Bandicota indica* Bechstein (Large bandicoot rat). *Bandicota indica* commonly inhabits cultivated fields, gardens, lawns and pasture lands. It mostly feeds on grains, succulent stems, roots, grasses, vegetables and fruits. Plenty of hairs are found to be present on the back of this rat. When disturbed it becomes furious, erects its hairs and starts grunting. It breeds throughout the year and the female has 10 mammae. This rat is largest of the field rats found in India and provided with short tail.

6. *Bandicota bengalensis bengalensis* Gray (Indian mole rat). This species commonly inhabits the fields of crops, vegetables and fruits. It is a fierce animal and grunts when in anger. This rat is very good swimmer. The dorsal surface of the rat is covered with thick harsh fur of grayish-brown colour and grayish-white on the ventral surface. Tail is somewhat shorter but sometimes as long as the body. The number of young ones per litter varies from 6 to 15. The burrows are generally shallower and have more than one opening. They stock the food grains inside their burrows for future feeding when food would not be available easily.

7. *Nesokia indica* Gray (Short tailed male rat). These mostly inhabit cultivated lands, lawns and gardens mainly feeding on field crops, vegetables, grass, fruits, etc. The body colour of this rat is brown dorsally and slightly pale on the ventral surface. *Nesokia indica* is found as a race of this species. The tail of this rat is shorter than the body length.

8. *Mus platythrix* Bennett (Brown spiny mouse). This is brown spiny mouse of common size. This rat burrows the side of the tank and so long as it remains inside, it keeps its opening closed with small pebbles. The body colour of this rat is dark-brown on dorsal surface while white on the ventral surface. Its tail is very short and it is always shorter than the body length. *Mus platythrix platythrix* (Bennett) occurs as a race of this species. The female has 10 mammae.

9. *Golunda ellioti* Gray (Indian bush rat). It inhabits the bushes or thorny branches or ground. This mostly feeds on vegetables and roots of grasses. Body colour of this rat is yellowish-brown on the dorsal surface and grey-brown on the ventral surface. *Golunda ellioti ellioti* Gray occurs as race of this species. The fur is coarse and not of spiny type. The tail of this rat is shorter than the

body length. It is provided with very small feet. This can breed throughout the year and the female has 8 mammae.

10. *Rattus meltdada* Gray (Soft fur field rat). This soft fur field rat inhabits cultivated fields and feeds on succulent plants, stems, grains and rhizomes of grasses. The fur is very soft and devoid of spines. The body colour of this rat is dark-brown or pale grey dorsally and greyish-white ventrally. *Rattus meltdada* Gray occurs as a race of this species. The colour of the tail is darker dorsally and pale from the ventral surface. This can breed throughout the year and the female is provided with 8 mammae.

11. *Millardia meltdada* Gray (Grass rat). This is found in wet lands as well as in dry lands and feeds on field crops, vegetables and fruits. This rat burrows in the field and bundhs. The tail is little shorter in length than the body. They may breed two times in a year and the female can produce 1 to 4 litters during breeding season. They keep 1 to 4 openings of the burrow and always have a few grasses in front of the opening of their burrows. They can breed throughout the year.

12. *Mus booduga* Gray (Common Indian field mouse). This is reported to inhabit in dry as well as wet lands and feeds on the grains of paddy. The colour is brownish on the dorsal surface and whitish or grey on the ventral surface. The tail is almost equal to the body length. The burrows formed by this rat are smaller, devoid of branching and provided with single bed chamber. They feed on sugarcane, maize and paddy. The female produces 6 to 13 young ones per litter.

Habit and Habitat

Rats are well adapted to adjust themselves everywhere because they are provided with highly developed sense of smell, taste, hearing and touch. Thus, they keep themselves away from any type of danger. They always go on their runways and remain careful and alert for any new object which they come across. They identify their routes with the help of dark greasy marks left in their way. In the night the rats usually come out of their burrows and actively keep themselves busy in feeding and in storage of food. At the time of this activity rats never get disturbed by any sound but as soon as some one reaches there they become alert. The home range of the males is greater than the female and varies from 50 to 100 feet from their burrows.

Formation of Burrow

Before the formation of burrows the rats survey the field, position of crop and humidity in the soil. The house rats dig their burrows in well sheltered places nearer to the store houses of grains, whereas, field rats form their burrows in the middle area of densely growing crops. They make a number

of openings (2 to 42) and chambers viz., breeding chamber and storage chambers in field burrows with a depth of 0.3 to 1.5 metre and diameter 0.03 to 0.685 metre. The length of the burrows is up to 40 metres with variations in the shape, size and depth-pattern according to the nature of rat species. The amount of food grains collected from a single burrow of *Bandicota* species was 0.035 to 5.94 Kg, highest of all other species. Thus, the main aim of the burrow formation is protection, favourable micro-climate, storage of food and breeding facilities. It is interesting that the temperature of burrows is lower in summer and higher in winter. After observing a large number of burrows it was found that six types of burrowing pattern exists :

- (1) Blind short with single opening.
- (2) Blind long with single opening.
- (3) Blind ramified with single opening.
- (4) U-type.
- (5) Ramified.
- (6) Intercommunicable.

The burrow's fauna and flora consist of earthworm, crickets, earwigs, beetles, ants, pseudoscorpion, millipede, mites etc.

Movements of Rats

Just after harvesting of crops, in unfavourable climatic conditions, due to shortage of food material in a particular area and other factors cause the migration of rats from one store to other or from one field to other. Generally in the months of July-August when fresh food materials are abundant in the field, rats (grass rat) come out of their burrows in search of suitable food grains in the night and damage paddy crops. The average distance travelled by grass rat was found to be 75.61 metres in 20 minutes. The maximum distance recorded at the time of migration is 551.0 metres in the field.

It has also been observed that rats are very good swimmers. The distance covered by swimming in five minutes has been recorded to be 16 to 19 metres in case of mole rat. The field rats having soft furs get exhausted quickly as compared to mole rats. The antelope rats are very poor swimmers but they were found climbing on trees and walls and can make horizontal jump of 8 feet.

Food and Nature of Demand

Commonly it is found that rats feed on all types of food materials viz., grains, fruits, vegetables, meat, fish, fish products, dairy products, etc., but they prefer fresh and clean food and some have taste pattern very much similar to that of human-beings. The order of food preference of the common rat is soaked wheat > soaked crushed barley > soaked lentils > soaked crushed maize > soaked millets > soaked rice. It has been observed from various data that the average consumption of grains by a rat is between 10 to 15 gm and maximum

grain consumption recorded is up to 30 gm per rat. It has also been noted that a pair of *Rattus melstada*, when given separately wheat, maize and grain for one week, consume 34.5, 39.3 and 38.7 gms respectively. But single rat feeds between 25.1 to 36.8 gm when given different food materials as mentioned above.

From a number of data it was generalised that rats eat about 10% of their body weight daily. They have sense to select balanced diet also. It is wonder that males can eat more amount of grains than the females in association with either themselves or with females.

The nature of damage by the rat is variable from species to species and crop to crop. The first attack of rats is on cereals to dicot seeds and grains. In wheat crop the maximum damage was recorded at the ripening stage. The loss calculated per hectare in the yield is about 52 Kg in the field in Punjab. Due to the alternative cropping scheme, the damage by rats per crop can be reduced easily. The damage of sugarcane crop due to rats in the field is much least in early stages but in later stages *i.e.*, from October, sugarcanes are attacked much by rats. In coconut fruits one or two holes are made by the rats near the stack and they damage the carpal of the coconut.

From the data available it is estimated that rat destroys 11.5% wheat, 5.8% barley and 0.9% gram in the field. They destroy 4.6% to 5.4% paddy, 4.1% to 25.8% ground nut, 1.4% maize, 2.2% sugarcane and 5% coconut.

It is interesting to note that the rats feed actively in 5 to 10 metres radius around their burrow. The amount of damage caused by rats is very much proportionate to the population of rats and their species in that locality.

The burrows made by rats weaken the foundation of buildings, cause seepage in channels, damage railway tracks in yards etc. They cause fire by damaging electric wires. The damage to the food material and other materials due to rats in U.K., France and U.S.A. has been estimated to be 15 million pounds, 15 million franks and 300 million dollars respectively.

Diseases Spread by Rats

Apart from damaging grains, fruits, vegetables, etc. rats spread a number of diseases among human-beings and their pets of common use. Thirty-two diseases of man and other animals are known to be transmitted by rats. They are reported to carry 18 different kinds of lice, fleas, ticks and mites. Different types of viruses, bacteria, fungi, protozoa, nematodes, arthropods cause several diseases like bubonic plague, typhus fever, infectious jaundice, rat bite fever and food-poisoning also. *Rattus rattus*, *Bandicota bengalensis* and *Tetra indica* are carriers of bubonic plague. A large number of ectoparasites are inhabiting the rats *viz.*, *Gleas (Xenopsylla cheopis, X. astia, Ctenocephalides felis)*, Mites

(*Dermanyssus muries*, *Laelaps echidninus*), Ticks (*R. sanguines*, *R. tricuspides*) and Louse (*Polyplax spinulosus*).

Life History

For the effective control measure an idea about the life history of rats is essential. After maturation, rats start living in separate burrows. In case of *Bandicota bengalensis bengalensis* the peak period for breeding is in two seasons i.e. March to May and August to October. After the age of 3 to 6 months they start breeding and gestation period lasts from 21 to 25 days. The female of this rat comes in heat after every 4 to 5 days for about 2 to 3 days and breeds commonly twice in a year coinciding with the maturity of the crops. The female produces 8 to 10 young ones in a litter but maximum number of young ones recorded is twenty two. Young ones produced are hairless, reddish and with closed eyes. After two weeks of birth, their eyes open and they begin to feed on grains after 3 weeks. The sex ratio is generally found to be 50 : 50. The life span of an adult female is longer than the male.

In case of *Millardia melitada melitada* the sex ratio of male and female is 5 : 9. The breeding season of grass rats in South India is November and January to April and the number of young ones per litter varies from 2 to 12. Thus, it is inferred that life history of rats may vary from species to species.

Ecological Status

The ecological study gives a clear picture about the interactions of rats with the nature. Cats, snakes, mongoose, owls and some other birds use the rats as their food. Thus, rats are also at important place in the ecological niche. Our aim should, therefore, not be to eradicate the rats completely but to reduce the rat population. However, very little is known about the ecology of the rats so the investigation on the ecology of the rats is needed for getting clear picture about them.

Rat Control ✓

Rat control is an important and complicated programme in which a number of control devices have been developed as result of practical experiences and scientific researches but none has succeeded in the control of rat population. The integrated control programme comprising of a number of control measures, has been highly recommended for the control of field and godown rats.

Methods of Rat Control

[I] Ultrasonic waves

Hanery Simon (1967) has recommended that ultrasonic waves are very much powerful killers of rats without any damage to human-beings. The wide use of ultrasonic waves in the field is under investigation specially with reference to its effect and high cost of this method for field use.

[II] Traps

This is an old method to catch the rats by various mechanical devices which are used in modified form in different localities and with the species of rat to be controlled. This method takes much time and man power so, practically very difficult to apply in larger field area. Sometimes this method is practised in houses to catch the house rats. The useful and common traps are bamboo bow and arrow trap and pot trap.

1. **Bamboo bow and arrow trap.** It is local made and very common in southern India. It comprises of triangular bamboo sticks fixed on bunds, tracks and burrows. As rats get in to take the food, the bamboo sticks snap automatically, trap and kill the rats.

2. **Pot trap.** It is a round earthen pot of 20 cm diameter and 17 cm height containing a lid and wood pegs of 30 cm. The lid is adjusted in such a way on the ground that the pot falls on the lid when disturbed. The bait is placed on the lid and also inside the pot. The bait when disturbed, falls on the lid and rat is trapped.

[III] Biological control

The natural enemies like cats, terrier, snakes, mongoose and a number of birds are predators of rats in the houses and the field. Cats are very common in houses and also go to the neighbouring fields in search of their prey. They catch the rats very actively. It is also found that in some offices cats are kept as pet specially for rats. Of all the predators snakes are traditionally regarded as the most effective predator of field rats and also of house rats. Some birds like kite, owl and hawk are active predators of rats in the field.

It is surprising that rats are eaten by human-beings also. In the 'Crush rats Campaign' organised in Indonesia (1964), Mr. D.N. Aidit was commander of that campaign. He suggested that the rats be consumed by human-beings.

[IV] Cultural control

Fitzwater and Prakash (1966) have suggested that change in the sowing and harvesting period of crops and the pattern of storage of grains may be quite helpful for rat control.

of vegetable oil (linseed, ground nut) should be mixed to make the food bait sticky. Now required quantity of Zinc-phosphide should be mixed with a long handled spoon. Then the required quantity of water should be added to make the pills in the following ratio :

Zinc-phosphide—1 part, Food bait—49 parts and vegetable oil—As required.

(b) *Barium carbonate*. Selected food bait is crushed and soaked in water for about 8 hours and then is allowed to dry and mixed with poison in following ratio :

Barium Carbonate—1 part, Food bait—6 parts and Jaggery—1 kg.

(c) *Red squill*. The dry food bait mixed with poison and required quantity of water is added to prepare the small balls. When fresh meat is used as food bait, a thin paste with poison and water is prepared and then mixed with the food bait as given below :

Red squill—1 part, Food bait—9 parts and water—as required.

(d) *Antu*. It is prepared in the same way as the red squill and gives most effective result. For the use of this, complete coverage of that area should be made and prepared by the formula as given below :

Antu—3 parts, Food bait—97 parts and water—as required.

(e) *Warfarin*. It is prepared similar to as Zinc phosphide. The difference is only in the preparation of baits and the formula is as given :

Warfarin—1 part, food—19 parts and vegetable oil—as required.

It is an anti-coagulant and has to be used for at least a week. Prakash (1976) has mentioned that poison fed rat suffers by weakness due to loss of blood and death may occur after 3 to 15 days of poisoning.

(f) *Tomorin*. It is prepared as above in case of warfarin with a slight variation in the proportion of bait and the poison as given :

Tomorin—1 part, Food bait—39 parts and vegetable oil—as required.

(g) *Sodium fluoroacetate*. Its use is restricted only to expert hands because it is highly toxic in nature for human-beings also. 1.5 ounce poison is mixed with one gallon of water and used as drinking bait.

(h) *Thallium sulphate*. This bait is prepared with wheat of rolled oat with or without oil. The poison should be mixed at the ratio of 1% by weight for achieving good results.

Precautions for the use of bait. (1) In residences, baits should be kept only at counted (2 to 3) places to avoid the risk of human hazards.

- (2) Generally 30 gm of bait is recommended in each living burrow but a slight change may be made according to the need.
- (3) The baiting should be done in clear weather.
- (4) The operation of baiting should be repeated till the burrows cease to be opened on next day.

- (5) The workers having cut on any portion of the hand or fingers should not be allowed to handle the poison baits.
- (6) The preparation of baits should be performed in a well ventilated room to avoid the lethal concentration of poisonous gas like, HCN or phosphine etc.
- (7) Baits should be kept 6 to 8 cm inside the burrows for proper placing of poison.
- (8) After handling the poison baits, hands should be cleaned by proper washing soap.
- (9) The nails of the hands should be cleaned and smaller as in usual course for a person.
- (10) The rats killed due to poison baiting should be buried in deep soil to avoid secondary poisoning of pets, poultry, predators and other useful animals.

Antidotes

Antidotes are the medicines that counteract the poisons. The discovery of antidotes in the field of chemical control measure has dawned a new era for the life of workers. The nature of antidotes varies with the kind of poisoning to counteract properly. It would be a good thing for human-beings if proper antidote can be prepared for every poison. Prakash (1976) has reported two different antidotes for zinc phosphide and warfarin. If poisoning has been caused due to zinc phosphide first of all mustard emetic should be given to induce vomiting. Six gm potassium permanganate, dissolved in a glass of water, should be given immediately. After 10 minutes one spoon copper sulphate, dissolved in 250 c.c. water should be given and it is followed by a purgative—a tea spoonfull of Epsom salt in glass of warm water.

In case of Warfarin (anticoagulant) first of all vitamin 'K' should be administered. Blood transfusion is also recommended. The physician should also be called at once.

Eradication of House and Godown Rats

Integrated approach by combining all control means in rat eradication campaign has proved very much fruitful. It is comprised of several steps like rat proofing, traps, cleanliness, sanitation and chemical approach.

1. Rat proofing. Rat proofing checks the movement of rats and mice into or outside a house. For this purpose all openings should be completely closed and the edges of all the doors and windows should be covered with metal sheets. Ventilators should be covered with 18 gauge 1.25 cm mesh. The sewer drains should be tightly fastened.

2. Traps. Traps are recommended only for small infestation. Cage trap, wooden trap, wire trap, snap trap are used to catch the rats. As soon as the rat

enters into the cage, the door of the cage is automatically closed. The food bait used may be of bread, biscuits, cakes, cheese, meat etc.

3. Cleanliness and sanitation. The food materials should not be kept open in tins or other vessels. The dump of refuse should be removed from main areas or should be burnt. There should be clean up programme in rat infested localities.

4. Chemicals. Eradication with chemical baits is more effective in godowns and houses. The whole programme consists of pre-baiting and application of poison baits.

In organising an anti-rat campaign in houses, farm houses and godowns following plan of operation should be followed. The persons should be given preliminary training regarding survey, marking of burrows and preparation of baits.

Plan for Operation

First day. Survey to locate run way and live burrows and prebaiting in the after noon.

Second day. Inspection of whole area and pre-baiting continued.

Third day. Poison baiting with zinc-phosphide, warfarin or Barium carbonate.

Fourth day. Re-examination of area and poison baiting in opened burrows with 2% zinc phosphide.

Fifth day. Collection of dead rats, baiting continued, if necessary fumigation with HCN may be done in alive burrows.

Sixth day. Use of chemosterilants if alive burrows are still found (Srivastava, 1968).

Eradication of Rat from Field

1. Trapping. Traps are not much effective in field and their application takes much time and specialized technique. Where baiting or fumigation is not possible traps may be used.

2. Chemicals. This programme comprises of pre-baiting, location of baits, application of poisoned baits, use of fumigants and chemosterilants.

First day. Survey of alive burrows and pre-baiting.

Second day. Pre-baiting continued.

Third day. Operation of poison baits as desired with regard to species of rats.

Fourth day. Collection of dead rats and burying in soil, continue baiting if necessary.

Fifth day. Re-examination of area, if necessary fumigation should be done.

Sixth day. If alive burrows still remain, use of chemosterilant is recommended, Srivastava (1968).

Success of rat eradication programme

For successful rat eradication programme, following care should be taken :

- (1) Rat control operation should be well planned and twice a year.
- (2) The eradication programme should be effective and economic.
- (3) The information about the biology, habitat and life cycle of a particular species of the rat of that field should be available.
- (4) The rat control programme should be operated in possibly larger areas.
- (5) The co-operation of local people becomes quite helpful, so local organisation should also be contacted.

Future of rat control

Literature regarding rat control reveals that although much work have been done in the laboratory condition but no hopeful success has been achieved in the field. It does not mean that results obtained from scientific works are not practicable in the field but this situation of failing in the 'Rat War' is only due to the careless steps of extension units and avoidance of farmers themselves. So, for the victory on this great enemy it is advisable for the Government to organise 'Rodent Control Exhibition' in villages in collaboration with the scientists working in this field and give them full training about each and every step of rat control. Professor Banett of Australian National University Canberra accepting the challenge of rats suggested to establish separate zoology department from Entomology department in all agricultural Universities. The Summer Institute Organised at Central Arid Zone Research Institute Jodhpur, by Indian Agriculture Research Institute has given some promising ideas for rat control programme. Separate Research Institute for study of the rats and their control would be of great help to the economy of the nation.